

Guia 6 E31

Movimiento de un Fluido con Componentes de Velocidad u y derivados de un Potencial ϕ

$$U = \frac{\partial \phi}{\partial x} \quad V = \frac{\partial \phi}{\partial y} \quad w = 0$$

$$a) \phi = \frac{1}{4\pi} \log(x^2 + y^2) = \frac{1}{2\pi} \log r \quad r^2 = x^2 + y^2$$

$$b) \phi = x$$

$$c) \phi = Ar^n \cos m\theta \quad \theta = \tan^{-1} \frac{y}{x}$$

$$d) \phi = \frac{\cos \theta}{r}$$

$$a) U = \frac{\partial \phi}{\partial x} \rightarrow U = \frac{\partial}{\partial x} \left(\frac{1}{4\pi} \log(x^2 + y^2) \right)$$

$$U = \frac{1}{4\pi} \cdot \frac{\partial}{\partial x} (\log(x^2 + y^2))$$

$$U = \frac{1}{4\pi} \cdot \frac{1}{(x^2 + y^2)} \cdot 2x$$

$$\left[U = \frac{x}{2\pi(x^2 + y^2)} \right]$$

$$V = \frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y} \left(\frac{1}{4\pi} \log(x^2 + y^2) \right)$$

$$V = \frac{1}{4\pi} \cdot \frac{1}{(x^2 + y^2)} \cdot 2y$$

$$\left[V = \frac{y}{2\pi(x^2 + y^2)} \right]$$

$$\begin{cases} \log u & u = x^2 + y^2 \\ \frac{1}{u} \cdot 2x = \frac{1}{x^2 + y^2} \cdot 2x \end{cases}$$

b)

$$U = \frac{\partial \phi}{\partial x} = \frac{\partial (x)}{\partial x}$$

$$U = 1$$

$$V = \frac{\partial \phi}{\partial y} \rightarrow \frac{\partial (x)}{\partial y}$$

$$V = 0$$

$$c) \phi = A r^m \cos m\theta$$

$$\phi(r, \theta) = \phi(x(r, \theta), y(r, \theta))$$

$$x = r \cos \theta \quad y = r \sin \theta$$

$$\frac{\partial \phi(r, \theta)}{\partial r} = \frac{\partial \phi(x, y)}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial \phi(x, y)}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\frac{\partial \phi(r, \theta)}{\partial r} = U_0 \cos \theta + V_0 \sin \theta$$

$$\frac{\partial \phi(r, \theta)}{\partial r} = U r$$

$$\frac{\partial \phi(r, \theta)}{\partial \theta} = \frac{\partial \phi(x, y)}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial \phi(x, y)}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$= -U_0 r \sin \theta + V_0 r \cos \theta$$

$$= r \cdot (V_0 \cos \theta - U_0 \sin \theta)$$

$$= r \cdot U \theta$$

$$\text{Entonces } U \theta = \frac{1}{r} \cdot \frac{\partial \phi(r, \theta)}{\partial \theta}$$

$$\text{Para } \phi = A \cdot r^m \cdot \cos(m\theta)$$

$$U r = \frac{\partial \phi}{\partial r} = A \cdot m \cdot r^{m-1} \cos(m\theta)$$

$$U \theta = \frac{1}{r} \cdot \frac{\partial \phi}{\partial \theta} = \frac{1}{r} \cdot A \cdot r^m \cdot (-\sin(m\theta)) m = A \cdot r^m \cdot r^{m-1} \cdot m \cdot \sin(m\theta)$$

$$U \theta = -A m \cdot r^{m-1} \cdot \sin(m\theta)$$

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$$d) \Phi = \frac{\cos \theta}{r} = \cos \theta \cdot r^{-1}$$

$$U_r = \frac{\partial \Phi}{\partial r} = \cos \theta \cdot (-1) \cdot r^{-2} = -\frac{\cos \theta}{r^2}$$

$$U_\theta = \frac{1}{r} \cdot \frac{\partial \Phi}{\partial \theta} = \frac{1}{r} \cdot \frac{1}{r} \cdot -\sin \theta = -\frac{\sin \theta}{r^2}$$

Exa 6 Ex 2

$$\Psi(r, \theta) = \Psi(x(r, \theta), y(r, \theta))$$

$$u = \frac{\partial \Psi}{\partial y}$$

$$v = -\frac{\partial \Psi}{\partial x}$$

$$\frac{d\Psi(r, \theta)}{d\theta} = \left[\frac{\partial \Psi(x, y)}{\partial x} \cdot \frac{dx}{d\theta} + \frac{\partial \Psi(x, y)}{\partial y} \cdot \frac{dy}{d\theta} \right]$$

$$\frac{d\Psi(r, \theta)}{d\theta} = \left[-v \cdot (-r \sin \theta) + u \cdot (r \cos \theta) \right]$$

$$\frac{d\Psi(r, \theta)}{d\theta} = r \cdot v \cdot \sin \theta + r \cdot u \cdot \cos \theta$$

$$= r \cdot (v \cdot \sin \theta + u \cdot \cos \theta)$$

$$\frac{d\Psi(r, \theta)}{d\theta} = r \cdot u_r$$

$$\left[u_r = \frac{1}{r} \frac{d\Psi(r, \theta)}{d\theta} \right]$$

$$\frac{d\Psi(r, \theta)}{dr} = \frac{\partial \Psi(x, y)}{\partial x} \cdot \frac{dx}{dr} + \frac{\partial \Psi(x, y)}{\partial y} \cdot \frac{dy}{dr}$$

$$\frac{d\Psi(r, \theta)}{dr} = -v \cdot \cos \theta + u \cdot \sin \theta$$
$$= -(v \cdot \cos \theta - u \cdot \sin \theta)$$

$$\frac{d\Psi(r, \theta)}{dr} = -u_\theta$$

$$u_\theta = -\frac{\partial \Psi(r, \theta)}{\partial r}$$

a) $\Psi = C \cdot \theta$

$$u_r = \frac{1}{r} \cdot C = \frac{C}{r}$$

$$u_\theta = 0$$

b) $\Psi = \gamma$

$$u = 1$$

$$v = 0$$

$$c) \psi = A_0 \cdot r^m \cdot \sin(m\theta)$$

$$U_r = \frac{1}{r} \cdot \frac{\partial \psi}{\partial \theta} = \frac{1}{r} \cdot A_0 \cdot r^m \cdot \cos(m\theta) \cdot m$$

$$U_r = A \cdot m \cdot r^{m-1} \cdot \cos(m\theta)$$

$$U_\theta = -\frac{\partial \psi}{\partial r} = -A \cdot m \cdot r^{m-1} \cdot \sin(m\theta)$$

$$d) \psi = -\frac{\sin \theta}{r} = -\sin \theta \cdot r^{-1}$$

$$U_r = \frac{1}{r} \cdot \left(-\frac{1}{r} \cdot \cos \theta \right) \quad U_r = -\frac{\cos \theta}{r^2}$$

$$U_\theta = -\frac{\partial \psi}{\partial r} = -(\sin \theta) \cdot (-1) \cdot r^{-2} = \frac{\sin \theta}{r^2}$$

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$$\text{Vorticidad: } \Omega = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} \left(\frac{\partial u_\theta}{\partial r} - \frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{u_\theta}{r} \right)$$

Mostrar que la vorticidad desaparece en cada caso

$$\text{a) } \left[\begin{array}{l} V = \frac{1}{2\pi} \cdot \frac{y}{(x^2+y^2)} \\ U = \frac{1}{2\pi} \cdot \frac{x}{(x^2+y^2)} \end{array} \right] \quad \begin{array}{l} \frac{\partial V}{\partial x} = \frac{1}{2\pi} \cdot y \cdot (-1) \cdot (x^2+y^2)^{-2} \cdot 2x \\ \frac{\partial V}{\partial y} = -\frac{1}{\pi} \cdot \frac{xy}{(x^2+y^2)^2} \end{array} \quad (1)$$

$$\frac{\partial U}{\partial y} = \frac{1}{2\pi} \cdot x \cdot (-1) \cdot (x^2+y^2)^{-2} \cdot 2y$$

$$\frac{\partial U}{\partial y} = -\frac{1}{\pi} \cdot \frac{xy}{(x^2+y^2)^2} \quad (2)$$

$$\Omega = \frac{1}{2} \cdot \left(-\frac{1}{\pi} \cdot \frac{xy}{(x^2+y^2)^2} - \left(-\frac{1}{\pi} \cdot \frac{xy}{(x^2+y^2)^2} \right) \right) = \frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

$$\text{b) } \begin{array}{l} V=0 \\ U=1 \end{array} \quad \begin{array}{l} \frac{\partial V}{\partial x} = 0 \\ \frac{\partial U}{\partial y} = 0 \end{array}$$

$$\Omega = \frac{1}{2} \cdot (0-0) = 0 \quad \checkmark$$

$$\text{c) } U_\theta = -A \cdot m \cdot r^{m-1} \cdot \ln(m\theta)$$

$$\frac{\partial U_\theta}{\partial r} = -m(m-1) \cdot A \cdot r^{m-2} \cdot \ln(m\theta)$$

$$U_r = A \cdot m \cdot r^{m-1} \cdot \cos(m\theta)$$

$$\frac{\partial U_r}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \sin(m\theta)$$

$$-\frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} = m^2 \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$\frac{U_\theta}{r} = -m \cdot A \cdot r^{m-2} \cdot \ln(m\theta)$$

$$\Omega = \frac{1}{2} \cdot \left(-m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \ln(m\theta) + m^2 \cdot A \cdot r^{m-2} \cdot \ln(m\theta) - m \cdot A \cdot r^{m-2} \cdot \ln(m\theta) \right)$$

$$\Omega = \frac{1}{2} \left((-m(m-1) + m^2 - m) \cdot A \cdot r^{m-2} \cdot \ln(m\theta) \right)$$

$$\Omega = \frac{1}{2} \underbrace{(-m^2 + m + m^2 - m)}_0 \cdot A \cdot r^{m-2} \cdot \ln(m\theta)$$

$$\Omega = \frac{1}{2} \cdot 0 = 0 \quad \checkmark$$

d)

$$U_\theta = -\frac{\ln \theta}{r^2}$$

$$U_r = -\frac{\cos \theta}{r^2}$$

$$-\frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta} = -\frac{\ln \theta}{r^3}$$

$$\frac{\partial U_\theta}{\partial r} = \frac{2 \cdot \ln \theta}{r^3}$$

$$\frac{\partial U_r}{\partial \theta} = -\frac{1}{r^2} \cdot (-\ln \theta)$$

$$\frac{U_\theta}{r} = -\frac{\ln \theta}{r^3}$$

$$\Omega = \frac{1}{2} \cdot \left(\frac{2 \ln \theta}{r^3} - \frac{\ln \theta}{r^3} - \frac{\ln \theta}{r^3} \right)$$

$$\Omega = \frac{1}{2} \cdot \left(\frac{2 \ln \theta}{r^3} - \frac{2 \ln \theta}{r^3} \right) = \Omega = \frac{1}{2} \cdot (0) = 0 \quad \checkmark$$

b) Expresiones para el Tensor Torzo de deformación

Tensor Torzo de deformación: $V_{ij} = V_{ji}$

$$V_{ij} = \frac{1}{2} \left(\frac{\partial V_j}{\partial x_i} + \frac{\partial V_i}{\partial x_j} \right) = \frac{1}{2} \left(\frac{\partial V}{\partial x} + \frac{\partial U}{\partial y} \right)$$

En coordenadas polares

$$V_{rr} = \frac{\partial U_r}{\partial r} \quad V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta} \quad V_{r\theta} = \left(\frac{1}{2} \cdot \left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r} \right) \right)$$

$$a) \frac{\partial V}{\partial x} = \frac{1}{2\pi} \left(\frac{x^2 + y^2 - x}{(x^2 + y^2)^2} \right)$$

$$\frac{\partial U}{\partial y} = \frac{1}{2\pi} \left(\frac{x^2 + y^2 - y}{(x^2 + y^2)^2} \right)$$

$$\left[V_{11} = \frac{\partial V}{\partial x} = \frac{1}{2\pi} \cdot \left(\frac{x^2 + y^2 - x}{(x^2 + y^2)^2} \right) \right] \quad V_{12} = \frac{1}{2} \left(-\frac{1}{\pi} \cdot \frac{x \cdot y}{(x^2 + y^2)^2} - \frac{1}{\pi} \cdot \frac{x \cdot y}{(x^2 + y^2)^2} \right)$$

$$\left[V_{22} = \frac{\partial U}{\partial y} = \frac{1}{2\pi} \cdot \left(\frac{x^2 + y^2 - y}{(x^2 + y^2)^2} \right) \right] \quad \left[V_{12} = -\frac{1}{\pi} \cdot \frac{x \cdot y}{(x^2 + y^2)^2} \right]$$

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b) $U=1$ $V=0$

$$\frac{\partial U}{\partial x} = 0$$

$$\frac{\partial V}{\partial y} = 0$$

$$\frac{\partial V}{\partial x} = 0$$

$$\frac{\partial U}{\partial y} = 0$$

$$V_{11} = \frac{\partial U}{\partial x} = 0$$

$$V_{12} = \frac{1}{2} (0 + 0) = 0$$

$$V_{22} = \frac{\partial V}{\partial y} = 0$$

c) $\frac{\partial U_r}{\partial r} = m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$

$$\frac{\partial U_\theta}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \cos(m\theta)$$

$$\frac{\partial U_\theta}{\partial r} = -m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$\frac{\partial U_r}{\partial \theta} = -m^2 \cdot A \cdot r^{m-1} \cdot \sin(m\theta)$$

$$V_{rr} = \frac{\partial U_r}{\partial r} = m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$$

$$V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta} = \frac{m \cdot A \cdot r^{m-1} \cdot \cos(m\theta)}{r} + \frac{1}{r} \cdot (-m^2 \cdot A \cdot r^{m-1} \cdot \cos(m\theta))$$

$$V_{\theta\theta} = m \cdot (1-m) \cdot A \cdot r^{m-2} \cdot \cos(m\theta)$$

$$V_{r\theta} = \frac{1}{2} \left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r} \right)$$

$$V_{r\theta} = m \cdot (m-1) \cdot A \cdot r^{m-2} \cdot \sin(m\theta)$$

$$d) \frac{\partial U_r}{\partial r} = \frac{2 \cdot \cos \theta}{r^3}$$

$$\frac{\partial U_\theta}{\partial \theta} = -\frac{\cos \theta}{r^2}$$

$$\frac{\partial U_\theta}{\partial r} = \frac{2 \sin \theta}{r^3}$$

$$\frac{\partial U_r}{\partial \theta} = \frac{\sin \theta}{r^2}$$

$$V_{rr} = \frac{\partial U_r}{\partial r} = \frac{2 \cos \theta}{r^3}$$

$$V_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta}$$

$$V_{\theta\theta} = -\frac{2 \cos \theta}{r^3}$$

$$V_{r\theta} = \frac{1}{2} \left(\frac{\partial U_\theta}{\partial r} + \frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} - \frac{U_\theta}{r} \right)$$

$$V_{r\theta} = \frac{2 \sin \theta}{r^3}$$

C) Verifier si les flux sont irrotationnels
 Rotationnel ou non

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$$U_1 = ax^2 + bxy + c \quad V = by^2 + cx + mz \quad W = mz^3$$

$$e_{ij} = \frac{1}{2} \left[\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right]$$

$$\begin{aligned} U_1 &= ax_1^2 + bx_1x_2 + c \\ U_2 &= bx_2^2 + cx_1 + mx_3 \\ U_3 &= mx_3^3 \end{aligned}$$

$$e_{11} = \frac{1}{2} \left[\frac{\partial U_1}{\partial x_1} + \frac{\partial U_1}{\partial x_1} \right] = \frac{1}{2} [2ax_1 + bx_2 + 2ax_1 + bx_2] = ax_1 + bx_2$$

$$e_{12} = \frac{1}{2} \left[\frac{\partial U_2}{\partial x_1} + \frac{\partial U_1}{\partial x_2} \right] = \frac{1}{2} [c + bx_1]$$

$$e_{13} = \frac{1}{2} \left[\frac{\partial U_3}{\partial x_1} + \frac{\partial U_1}{\partial x_3} \right] = \frac{1}{2} [0 + 0] = 0$$

$$e_{21} = \frac{1}{2} \left[\frac{\partial U_1}{\partial x_2} + \frac{\partial U_2}{\partial x_1} \right] = \frac{1}{2} [bx_1 + c]$$

$$e_{22} = \frac{1}{2} \left[\frac{\partial U_2}{\partial x_2} + \frac{\partial U_2}{\partial x_2} \right] = \frac{1}{2} [2bx_2 + 2bx_2] = \frac{1}{2} \cdot 2(bx_2 + bx_2) = 2bx_2$$

$$e_{23} = \frac{1}{2} \left[\frac{\partial U_3}{\partial x_2} + \frac{\partial U_2}{\partial x_3} \right] = \frac{1}{2} [0 + m] = \frac{1}{2}m$$

$$e_{31} = \frac{1}{2} \left[\frac{\partial U_1}{\partial x_3} + \frac{\partial U_3}{\partial x_1} \right] = \frac{1}{2} [0 + 0] = 0$$

$$e_{32} = \frac{1}{2} \left[\frac{\partial U_2}{\partial x_3} + \frac{\partial U_3}{\partial x_2} \right] = \frac{1}{2} [m + 0] = \frac{1}{2}m$$

$$e_{33} = \frac{1}{2} \left[\frac{\partial U_3}{\partial x_3} + \frac{\partial U_3}{\partial x_3} \right] = \frac{1}{2} [3mx_3^2 + 3mx_3^2] = \frac{1}{2} \cdot 3(mx_3^2 + mx_3^2) = \frac{3}{2} \cdot 2(mx_3^2) = 3mx_3^2$$

$$e_{33} = 3mx_3^2$$

$$\frac{\partial^2 e_{11}}{\partial x_1 \partial x_3} = 0$$

$$\frac{\partial}{\partial x_1} \left(-\frac{\partial e_{23}}{\partial x_1} + \frac{\partial e_{32}}{\partial x_2} + \frac{\partial e_{12}}{\partial x_3} \right)$$

$$\frac{\partial}{\partial x_1} (-0 + 0 + 0) = 0$$

$$[0 = 0]$$



$$\frac{\partial^2 e_{12}}{\partial X_3 \partial X_1} = \frac{\partial}{\partial X_3} \cdot \frac{\partial e_{12}}{\partial X_1} = \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_1} (2bX_2) = \frac{\partial}{\partial X_3} (0) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_2} \left(-\frac{\partial e_{31}}{\partial X_2} + \frac{\partial e_{12}}{\partial X_3} + \frac{\partial e_{23}}{\partial X_1} \right) = \frac{\partial}{\partial X_2} (-0 + 0 + 0) = 0 \quad (2)$$

$$(1) = (2)$$

$$0 = 0 \checkmark$$

$$\frac{\partial^2 e_{32}}{\partial X_1 \partial X_2} = \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2} (3mX_3^2) = \frac{\partial}{\partial X_1} (0) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_3} \left(-\frac{\partial e_{12}}{\partial X_3} + \frac{\partial e_{23}}{\partial X_1} + \frac{\partial e_{31}}{\partial X_2} \right) = \frac{\partial}{\partial X_3} (0 + 0 + 0) = 0 \quad (2)$$

$$(1) = (2)$$

$$0 = 0 \checkmark$$

$$2 \cdot \frac{\partial}{\partial X_1} \cdot \frac{\partial e_{33}}{\partial X_2} = 2 \cdot \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2} (3mX_3^2) = 2 \cdot \frac{\partial}{\partial X_1} (0) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_2} \cdot \frac{\partial}{\partial X_2} (2X_1 + bX_2) + \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_1} (2bX_2) = \frac{\partial}{\partial X_2} (b) + \frac{\partial}{\partial X_1} (0) = 0 \quad (2)$$

$$(1) = (2) = 0 \checkmark$$

$$2 \cdot \frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_2} \left(\frac{1}{2} m \right) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_3} (2bX_2) + \frac{\partial}{\partial X_2} \cdot \frac{\partial}{\partial X_2} (3mX_3^2) = 0 + 0 = 0 \quad (2) \quad (1) = (2) = 0 \checkmark$$

$$2 \cdot \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_1} (0) = 0 \quad (1)$$

$$\frac{\partial}{\partial X_1} \cdot \frac{\partial}{\partial X_1} (3mX_3^2) + \frac{\partial}{\partial X_3} \cdot \frac{\partial}{\partial X_3} (2bX_2) = 0 \quad (2) \quad 1 = 2 = 0 \checkmark$$

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$$U = a \cdot r \cdot \log(\theta) \quad V = a \cdot r^2 + C \cdot \ln(\theta) \quad W = 0$$

$$U_r = U \cdot \cos \theta + V \cdot \ln \theta$$

$$U_r = a \cdot r \cdot \log(\theta) \cdot \cos(\theta) + a \cdot r^2 \cdot \ln(\theta) + C \cdot \ln^2(\theta)$$

$$U_\theta = -U \cdot \ln(\theta) + V \cdot \cos(\theta)$$

$$U_\theta = -a \cdot r \cdot \log(\theta) \cdot \ln(\theta) + a \cdot r^2 \cdot \cos(\theta) + C \cdot \ln(\theta) \cdot \cos(\theta)$$

El campo de desplazamientos dado por U, V y W es continuo y diferenciable

En la componente U , $\log(\theta)$ es discontinuo para $\theta = 0$

Se reduce el problema a $0 < \theta < 2\pi$ para evitar la discontinuidad y para que el campo sea real. Como el campo es continuo y diferenciable es compatible

Considerando el campo de desplazamientos en coordenadas polares:

$$e_{rr} = \frac{\partial U_r}{\partial r} \quad e_{\theta\theta} = \frac{U_r}{r} + \frac{1}{r} \cdot \frac{\partial U_\theta}{\partial \theta}$$

$$e_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \cdot \frac{\partial U_r}{\partial \theta} + \frac{\partial U_\theta}{\partial r} - \frac{U_\theta}{r} \right) \quad e_{zr} = \frac{1}{2} \left(\frac{\partial U_r}{\partial z} + \frac{\partial U_z}{\partial r} \right)$$

$$e_{z\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial U_z}{\partial \theta} + \frac{\partial U_\theta}{\partial z} \right) \quad e_{zz} = \frac{\partial U_z}{\partial z}$$

$$\text{Donde } U(r, \theta) = \begin{bmatrix} U_r \\ U_\theta \\ U_z \end{bmatrix} = \begin{bmatrix} U_r(r, \theta) \\ U_\theta(r, \theta) \\ U_z(r, \theta) \end{bmatrix}$$

Verificamos el campo que es compatible con la expresion S_{zz} en $\sigma_{\theta\theta}$

Gua 6 Ej 6

a) Demuestra que el vector de rotación puede interpretarse físicamente como la rotación que sufre el entorno del punto considerado

Tenemos $\partial U_i = \frac{\partial U_i}{\partial X_j} \partial X_j$

$$\begin{aligned} & \frac{1}{2} \frac{\partial U_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial U_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial U_j}{\partial X_i} \partial X_j - \frac{1}{2} \frac{\partial U_i}{\partial X_j} \partial X_j \\ &= \frac{1}{2} \left(\frac{\partial U_i}{\partial X_j} - \frac{\partial U_j}{\partial X_i} \right) \partial X_j + \frac{1}{2} \left(\frac{\partial U_j}{\partial X_i} + \frac{\partial U_i}{\partial X_j} \right) \partial X_j \\ &= \underbrace{\frac{1}{2} \left(\frac{\partial U_i}{\partial X_j} - \frac{\partial U_j}{\partial X_i} \right)}_{W_{ij}} \partial X_j \\ &= -W_{ij} \partial X_j + \epsilon_{ij} \partial X_j \\ &= -W_{ij} \partial X_j \\ &= -\epsilon_{ijk} W_k \partial X_j \\ &= \epsilon_{ikj} W_k \partial X_j = (W \times \partial X)_i \end{aligned}$$

$$b) W_k = \frac{1}{2} \sum_{ij} \epsilon_{kij} W_{ij} \rightarrow W_{ij} = \epsilon_{ijk} W_k$$

$$W_{ij} = \epsilon_{ijk} W_k \quad \text{multiplico ambos lados}$$

$$\epsilon_{ijp} W_{ij} = \epsilon_{ijp} \epsilon_{ijk} W_k \quad \text{Reordenar de } \epsilon_{ijk}$$

$$\epsilon_{ijp} W_{ij} = (\epsilon_{ijk} \epsilon_{ijp}) W_k \quad S_{ij} = 3 \text{ y contra } \epsilon_{ijk}$$

$$\epsilon_{ijp} W_{ij} = (\delta_{kp} - 3\delta_{pk}) W_k \quad \text{distingo y cambio } S_{pk} \text{ a } S_{kp}$$

$$\epsilon_{ijp} W_{ij} = \underline{S_{kp} W_k} - 3 \underline{S_{kp} W_k} \quad \text{contra } \epsilon_{ijk}$$

$$\epsilon_{ijp} W_{ij} = W_p - 3W_p$$

$$\epsilon_{ijp} W_{ij} = -2W_p$$

$$-\frac{1}{2} \epsilon_{ijp} W_{ij} = W_p \quad \text{reordenar } W_{ij} = W_{ji}$$

$$-\frac{1}{2} \epsilon_{pji} W_{ji} = W_p$$

$$\frac{1}{2} \epsilon_{pji} W_{ji} = W_p \quad \text{contra } \epsilon_{ijk} \text{ para que quede igual que el anterior}$$

$$\left[\frac{1}{2} \sum_{kij} \epsilon_{kij} W_{ij} = W_k \right]$$

Guia 6 E36

$$c) \Omega_k = \epsilon_{kij} \Omega_{ij} \rightarrow \Omega_{ij} = \frac{1}{2} \epsilon_{ijk} \Omega_k$$

$$\Omega_{ij} = \frac{1}{2} \epsilon_{ijk} \Omega_k \quad \text{diferencia de vector de rotación}$$

$$= \frac{1}{2} \epsilon_{ijk} \epsilon_{k\ell m} \Omega_{\ell m}$$

$$= \frac{1}{2} \epsilon_{kij} \epsilon_{k\ell m} \Omega_{\ell m} \quad \begin{array}{l} \text{Resolvemos índices} \\ \text{Relación delta-Epsilon} \end{array}$$

$$= \frac{1}{2} (\delta_{ie} \delta_{jm} - \delta_{im} \delta_{je}) \Omega_{\ell m} \quad \text{distributiva}$$

$$= \frac{1}{2} (\underbrace{\delta_{ie} \delta_{jm} \Omega_{\ell m}}_{\text{Contrahemos índices}} - \underbrace{\delta_{im} \delta_{je} \Omega_{\ell m}}_{\text{Contrahemos índices}})$$

$$= \frac{1}{2} (\Omega_{ij} - \Omega_{ji}) \quad \text{Por antisimetría del tensor de rotación}$$

$$= \frac{1}{2} (\Omega_{ij} + \Omega_{ij})$$

$$= \frac{1}{2} \cdot 2 \Omega_{ij}$$

$$= \Omega_{ij}$$

d) Demonstrar $\Omega = \nabla \times V$

$$\nabla \times V = \epsilon_{ijk} \frac{\partial V_k}{\partial x_j}$$

$$\nabla \times V = \epsilon_{ijk} \left(\frac{1}{2} \frac{\partial V_k}{\partial x_j} + \frac{1}{2} \frac{\partial V_k}{\partial x_j} + \frac{1}{2} \frac{\partial V_j}{\partial x_k} - \frac{1}{2} \frac{\partial V_j}{\partial x_k} \right)$$

$$\nabla \times V = \epsilon_{ijk} \left(\frac{1}{2} \frac{\partial V_k}{\partial x_j} - \frac{1}{2} \frac{\partial V_j}{\partial x_k} \right) + \epsilon_{ijk} \frac{1}{2} \left(\frac{\partial V_k}{\partial x_j} + \frac{\partial V_j}{\partial x_k} \right)$$

Reconhecer y m d i n g s
diferença Ω

$$\nabla \times V = \epsilon_{ijk} \Omega_k + \frac{1}{2} \left(\epsilon_{ijk} \frac{\partial V_k}{\partial x_j} + \epsilon_{ijk} \frac{\partial V_j}{\partial x_k} \right)$$

Combinar de ϵ_{ijk} e ϵ_{ikj}

$$\nabla \times V = \epsilon_{ijk} \Omega_k + \frac{1}{2} \left(\epsilon_{ijk} \frac{\partial V_k}{\partial x_j} + \epsilon_{ikj} \frac{\partial V_k}{\partial x_j} \right)$$

$$\nabla \times V = \epsilon_{ijk} \Omega_k + \frac{1}{2} \left(\epsilon_{ijk} \frac{\partial V_k}{\partial x_j} - \epsilon_{ijk} \frac{\partial V_k}{\partial x_j} \right)$$

$$\textcircled{1} = \textcircled{2} \text{ cancela}$$

$$\textcircled{1} - \textcircled{2} = 0$$

$$\nabla \times V = \epsilon_{ijk} \Omega_k + \frac{1}{2} \cdot 0$$

$$\nabla \times V = \epsilon_{ijk} \Omega_k \quad [\Omega_k = \epsilon_{kij} \Omega_{ij}]$$

$$\nabla \times V = \Omega_i$$

$$\nabla \times V = \Omega$$

$$e) \partial V_i = \frac{\partial V_i}{\partial X_j} \partial X_j$$

$$= \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_i}{\partial V_j} \partial X_j$$

$$= \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_i}{\partial X_j} \partial X_j + \frac{1}{2} \frac{\partial V_j}{\partial X_i} \partial X_j - \frac{1}{2} \frac{\partial V_j}{\partial X_i} \partial X_j$$

$$= \frac{1}{2} \left(\frac{\partial V_i}{\partial X_j} - \frac{\partial V_j}{\partial X_i} \right) \partial X_j + \frac{1}{2} \left(\frac{\partial V_j}{\partial X_i} + \frac{\partial V_i}{\partial X_j} \right) \partial X_j$$

$$= -\Omega_{ij} \partial X_j + V_{ij} \partial X_j$$

$$- \Omega_{ij} \partial X_j$$

$$- \frac{1}{2} \varepsilon_{ijk} \Omega_k \partial X_j$$

$$\frac{1}{2} \varepsilon_{ijk} \Omega_k \partial X_j$$

$$\frac{1}{2} (\Omega \times \partial X)_i$$

EJERCICIOS EN OCTAVE

Ejercicio 1 y 2

```
clc;

[x,y]=meshgrid(-4:1:4,-4:1:4);

% --- 1 ---
%---ej 1
u1 = (1/(2*pi))*(x./(x.^2+y.^2));
v1 = (1/(2*pi))*(y./(x.^2+y.^2));
figure(1);
quiver(x,y,u1,v1,'b');
hold on;
%----ej 2
% c = 0;
% u1 = (c/r) = 0/r = 0;
u2 = zeros(size(x));
% v1 = 0;
v2 = zeros(size(x))
quiver(x,y,u2,v2,'r')
title('Potencial y Corriente 1');
axis tight;
xlabel("x");
ylabel("y");

%--- 2 ---
%----ej 1
u2 = 1;
u2 = ones(length(x),length(y));
v2 = 0;
v2 = zeros(length(x),length(y));
figure(2);
quiver(x,y,u2,v2,'b');
hold on;
%----ej 2
u2 = ones(length(x),length(y));
v2 = zeros(length(x),length(y));
quiver(x,y,u2,v2,'r')
title('Potencial y Corriente 2');
axis tight;

% --- 3 ---
%---ej1
%definicion de la funcion
```

```

A = 1;
n = 2;
r = x.^(2) + y.^(2);
theta = atan2(y,x);
u_r = A*n*(r.^(n-1))*cos(n*theta);
u_theta = -A*n*(r.^(n-1))*sin(n*theta);
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
figure 3;
quiver(x,y,u,v,'b');
hold on
title('Potencial y Corriente 3');
axis tight;

%---ej2
u_r = A*n*(r.^(n-1))*cos(n*theta);
u_theta = -A*n*(r.^(n-1))*sin(n*theta);

u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
quiver(x,y,u,v,'r');

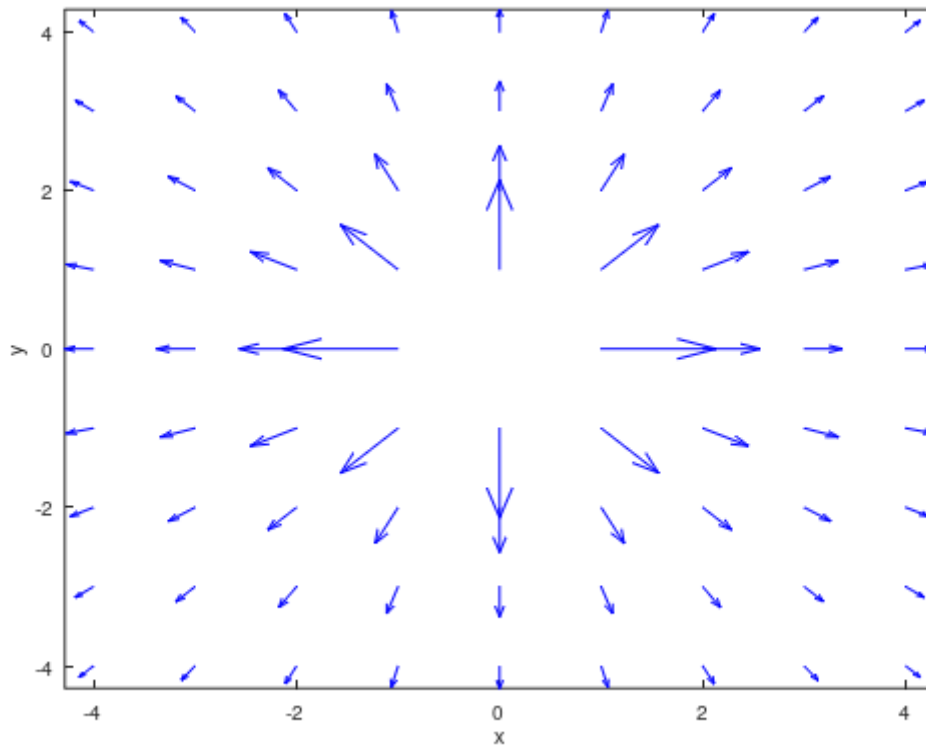
## % --- 4 ---
%---ej1
u_r = -cos(theta)./(r^(2));
u_theta = -sin(theta)./(r^(2));
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
figure 4;
quiver(x,y,u,v,'b');
hold on;
title('Potencial y Corriente 4');
axis tight;

%--ej2
u_r = -cos(theta)./(r^(2));
u_theta = -sin(theta)./(r^(2));
u = u_r*cos(theta) - u_theta*sin(theta);
v = u_r*sin(theta) + u_theta*cos(theta);
quiver(x,y,u,v,'r');

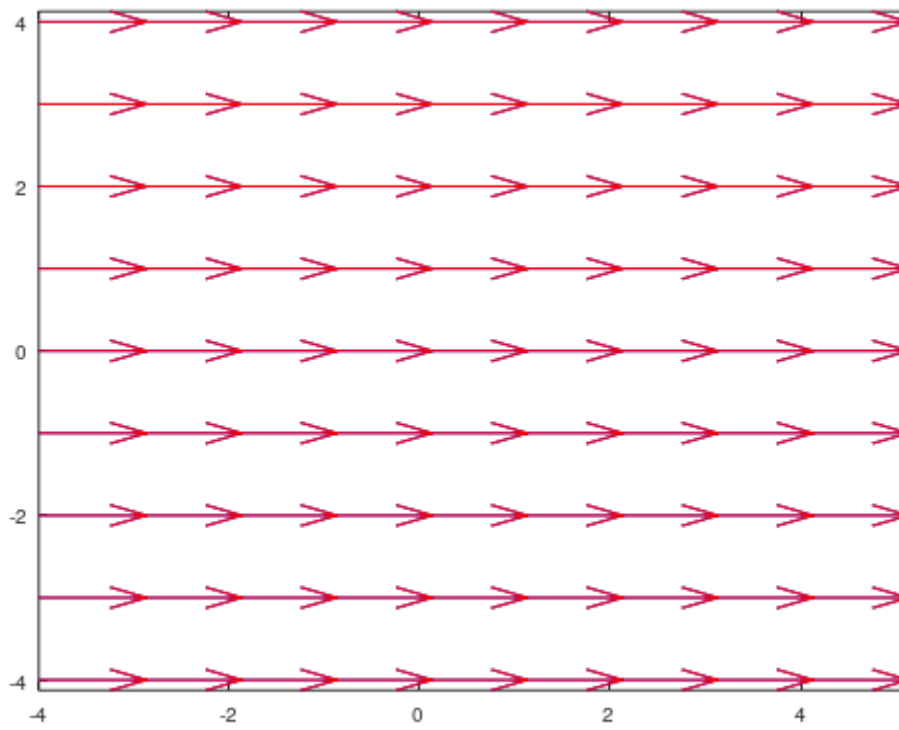
```

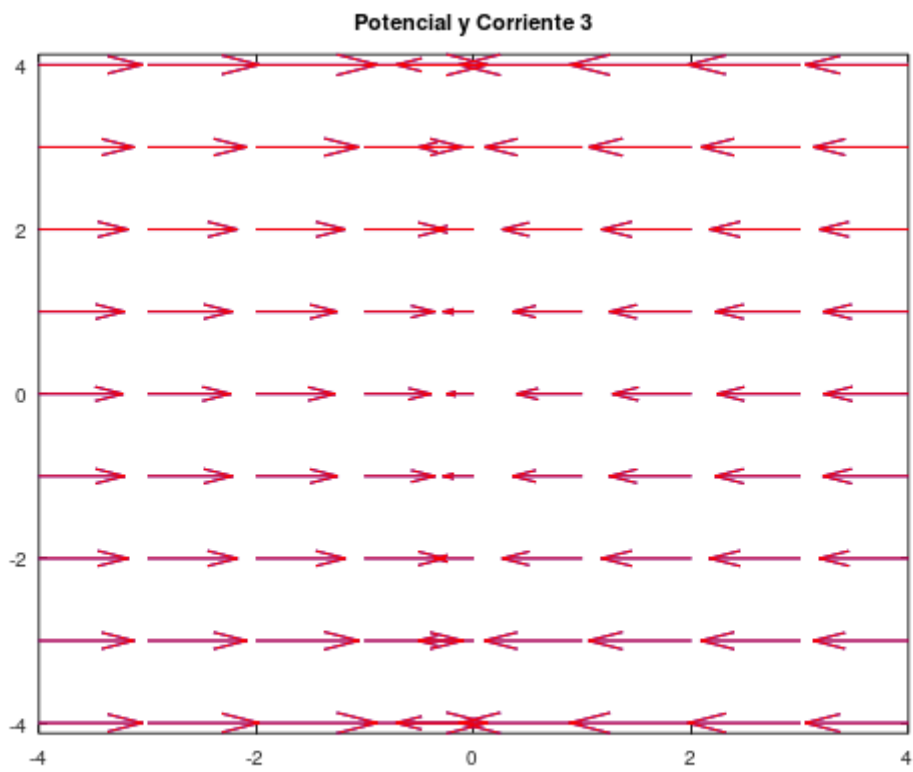
Graficas de potencial y corriente

Potencial y Corriente 1

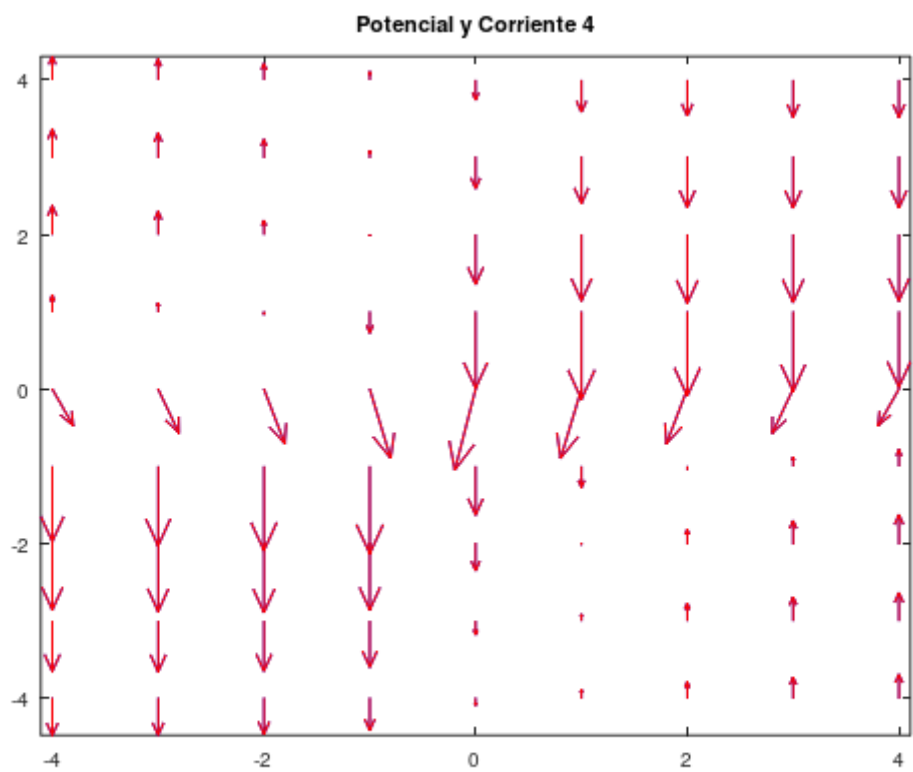


Potencial y Corriente 2





$(-3.941, 3.7168)$



$(-3.3471, 4.2839)$

Ejercicio 3 c)

..

```

pkg load symbolic;

% Verificación de divergencia:
% Si  $\text{div}(\mathbf{v}) = 0$ , entonces el flujo es incompresible

syms x y r theta A n c

% Potenciales y sus divergencias
phi_a = (1/(2*pi))*log(r); % Calculado utilizando coordenadas polares
div_phi_a = diff(diff(phi_a, r), r) + (1/(r^2))*diff(diff(phi_a, theta), theta) + (1/r)*(diff(phi_a, r));

phi_b = x;
div_phi_b = diff(diff(phi_b, x), x) + diff(diff(phi_b, y), y);

phi_c = A*(r^n)*cos(n*theta);
div_phi_c = diff(diff(phi_c, r), r) + (1/(r^2))*diff(diff(phi_c, theta), theta) + (1/r)*(diff(phi_c, r));

phi_d = cos(theta)/r;
div_phi_d = diff(diff(phi_d, r), r) + (1/(r^2))*diff(diff(phi_d, theta), theta) + (1/r)*(diff(phi_d, r));

% Funciones de corriente y sus divergencias
psi_a = c*theta;
div_psi_a = diff(diff(psi_a, r), r) + (1/(r^2))*diff(diff(psi_a, theta), theta) + (1/r)*(diff(psi_a, r));

psi_b = y;
div_psi_b = diff(diff(psi_b, x), x) + diff(diff(psi_b, y), y);

psi_c = A*(r^n)*sin(n*theta);
div_psi_c = diff(diff(psi_c, r), r) + (1/(r^2))*diff(diff(psi_c, theta), theta) + (1/r)*(diff(psi_c, r));

psi_d = -sin(theta)/r;
div_psi_d = diff(diff(psi_d, r), r) + (1/(r^2))*diff(diff(psi_d, theta), theta) + (1/r)*(diff(psi_d, r));

% Mostrar los resultados
disp('div_phi_a =');
disp(div_phi_a);

disp('div_phi_b =');
disp(div_phi_b);

disp('div_phi_c =');
disp(div_phi_c);

```

```

disp('div_phi_d =');
disp(div_phi_d);

disp('div_psi_a =');
disp(div_psi_a);

disp('div_psi_b =');
disp(div_psi_b);

disp('div_psi_c =');
disp(div_psi_c);

disp('div_psi_d =');
disp(div_psi_d);

```

Ejercicio 5

```

pkg load symbolic;
clear;
close all;

%-----
%----Inciso a---
%-----

% Declaración de variables
syms a r theta c z;
syms e_r_r e_theta_theta e_r_theta e_z_r e_z_theta e_z_z;

% Campo de desplazamiento
u_r = a*r*log(theta);
u_theta = a*r^2 + c*sin(theta);
u_z = 0;

% Componentes del tensor
e_r_r = diff(u_r,r);
e_theta_theta = (u_r / r) + (1 / r)*diff(u_theta,theta);
e_r_theta = (1/2)*((1/r)*diff(u_r,theta) + diff(u_theta,r) - (u_theta / r));
e_z_r = (1/2)*(diff(u_r,z) + diff(u_z,r));
e_z_theta = (1/2)*((1/r)*diff(u_z,theta) + diff(u_theta,z));
e_z_z = diff(u_z,z);

% Verificación de compatibilidad
szz = (2/r)*diff(diff(e_r_theta,theta),r) -...
      diff(diff(e_theta_theta,r),r) -...
      (1/(r^2))*diff(diff(e_r_r,theta),theta) +...
      (1/r)*diff(e_r_r,r) +...

```

```

        (2/(r^2))*diff(e_r_theta,theta) -...
        (2/r)*diff(e_theta_theta,r);

f_szz = function_handle(szz);

a = 1;
c = 1;
r = 1;

theta = linspace(1,pi/2,2*pi-1); % 0 < theta < 2*pi
valor_szz = f_szz(a,c,r,theta) % Debe ser 0 para que el campo sea compatible

%---Inciso b---
%-----
% Gráfico del campo de desplazamiento

rango = linspace(-3,3,50);
[x,y] = meshgrid(rango,rango);
r = x.^2 + y.^2;
theta = atan2(y,x);
a = 1;
c = 1;

u = a*r*log(theta)*cos(theta) + a*(r.^2)*sin(theta) + c*(sin(theta).^2);
v = -a*r*log(theta)*sin(theta) + a*(r.^2)*cos(theta) + c*sin(theta)*cos(theta);

% Gráfico
figure(1);
quiver(x,y,u,v);
title('Campo de Desplazamiento');
xlabel("x");
ylabel("y");
axis tight;

```

Campo de desplazamiento ejercicio 5

Campo de Desplazamiento

